2.2 Transport and gas exchange systems

This section provides learners with a knowledge and understanding of the structure and functions of the transport and gas exchange systems in mammals and multicellular terrestrial plants.

Learners will gain an appreciation of the need for mass transport systems and the significance of the surface area to volume ratio in multicellular organisms. The mass flow of materials within mammals and the exchange of molecules are studied in the context of the need to maintain a high metabolic rate.

Technological advances enable the activity of the heart, circulation and gas exchange systems to be monitored. Learners will be able to apply their knowledge of physiology to the interpretation of data on vital measurements of heart function, blood pressure and pulmonary ventilation. The importance of timely emergency treatment in both cardiac and respiratory arrest is also studied.

In this section, learners gain an understanding of the transport systems of terrestrial plants. In crop plants, the sites where organic molecules are stored – the harvestable parts of the plant – are often different to the sites where these molecules are produced – the leaves. This means that these molecules must be moved around the plant as required. The movement of water through a plant is, however, mainly due to physical mechanisms and processes. Learners will investigate the transport systems of multicellular plants and the different tissues involved. They will gain an appreciation of how the distribution of these tissues varies between different types of plants and that this distribution is diagnostic of the type of crop plant (broad-leaved or cereal) and the structural part of the plant (stem or root).

Learners are expected to apply knowledge, understanding and other skills gained in this section to new situations and/or to solve related problems.